

**Credit Name:** CSE 2110 Procedural Programming 1  
**Assignment Name:** DigitsDisplay

## Error Log Entry

### What error message did you encounter (if any)?

Error message that I encountered during my code was that I did not know how to display my code from the highest digit to the lowest digit. It was doing it the reverse order. Another error message that I encountered was that I had a syntax error with my do-while loop.

### What unexpected behavior did your program exhibit?

An unexpected error that my program encountered was that my numbers were displaying from reverse order, not highest digit to lowest digit.

### What caused the issue? (e.g., syntax error, logic error, incorrect function usage, etc.)

The do-while loop was a syntax error. The code displaying the numbers reverse order was a logic error.

### Include a screenshot of specific lines of code.

```
do
{
    System.out.print("Enter a positive integer: ");
    a = input.nextInt();

    if (a < 0)
    {
        System.out.println("You did not enter a positive integer. Please enter a positive integer.");
    }
}
while (a < 0);

while (a > 0)
{
    int display = a % 10;
    System.out.println(display);
    a /= 10;
}
}
```

### How did you fix the issue?

I was able to fix the do-while loop by ending the do with a curly bracket and then underneath it I put a while condition. This fixed the syntax error and my do-while loop started to work. Furthermore, to fix my issue with the code displaying from the highest digit to lowest digit, I started to make different variables. I made a variable called divisor and another variable called temp. With the divisor, I made it do all the important calculations that would make the code display it. With temp, I created it where the number can keep referring to it when it needs to without changing the original number.

Provide the corrected code or solution using a screenshot.

```
while (a < 0); // Repeat the loop if the number is negative

int divisor = 1; // Will store the highest place value divisor (1, 10, 100, etc.)
int temp = a;    // Temporary copy of the number so we don't modify the original

// Determine the largest place value of the number
// Example: if a = 5432, divisor becomes 1000
while (temp >= 10)
{
    temp /= 10;    // Remove the last digit from temp
    divisor *= 10; // Increase divisor to the next place value
}

// Loop to display each digit from left to right
while (divisor > 0)
{
    int display = a / divisor; // Get the leftmost digit
    System.out.println(display); // Print the digit on a new line

    a %= divisor; // Remove the printed digit from the number
    divisor /= 10; // Move to the next place value (1000 → 100 → 10 → 1)
}
```

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